# Blessed G/Michael Catholic Secondary School

## Semester 1 Physics Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

**Name:** _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

**I. Write 'True' if the statement is correct and 'False' if it is incorrect.**

____ **1.** The ability to control exhalation allows long phrases to be sung in a controlled and relaxed manner.

____ **2.** Hyper-echoic areas appear black on ultrasound because they do not send back any sound waves.

____ **3.** The direction of angular acceleration is in the same direction as the direction of angular velocity if the angular speed is decreasing in time.

____ **4.** When a wheel of radius R rotates about a fixed axis, all points on the wheel have the same angular speed.

____ **5.** According to Kepler's laws of planetary motion, a satellite increases its speed as it approaches the Sun and decreases its speed as it moves away from the Sun.

____ **6.** The gravitational force between a 60 kg man and Earth is not equal because Earth is more massive than the man therefore, it exerts the greatest force.

____ **7.** A sealed empty plastic water bottle collapses as an airplane descends from a higher elevation to a lower elevation.

____ **8.** As a fluid's speed increases, the pressure that the moving fluid exerts on the surface decreases.

____ **9.** The magnetic field B at a distance r from a long straight wire carrying current I is directly proportional to r.

____ **10.** In Lenz's law the induced emf opposes the change in magnetic flux.

____ **11.** An electromagnet loses its magnetism when the electric current is off.

____ **12.** The magnetic field lines are denser where the magnetic field is stronger.

**II. Choose the best answer among the given alternatives.**

____ **13.** A rectangular wooden block floats with 75% of its volume inside water. What is the density of this block? (density of water = 1000 kg/m³)

**A.** 75000 kg/m³ **B.** 750 kg/m³ **C.** 25000 kg/m³ **D.** 250 kg/m³

____ **14.** How would the period of an object in a circular orbit change if the radius of the orbit doubled?

**A.** The period would increase by a factor of 2.
**B.** The period would decrease by a factor of 4.
**C.** The period would increase by a factor of 2√2.
**D.** The period would decrease by a factor of 2√2.

____ **15.** If you double the mass of a certain substance, what will happen to its density?

**A.** It will be doubled. **B.** It will be halved.
**C.** It remains the same. **D.** It is indeterminate.

____ **16.** A fluid that occupies a volume of 20 L weighs 240 N at a location where the gravitational acceleration is 10 m/s². What is the mass of this fluid?

**A.** 20 kg **B.** 24 kg **C.** 40 kg **D.** 220 kg

____ **17.** An incompressible fluid flows along a pipe of decreasing radius (cross-sectional area decreases from A1 to A2). Which one of the following is necessarily true?

**A.** The flow rate must be increasing along the pipe of decreasing radius.
**B.** The flow rate must be decreasing along the pipe of decreasing radius.
**C.** The fluid pressure must be increasing along the pipe of decreasing radius.
**D.** The fluid pressure must be decreasing along the pipe of decreasing radius.

____ **18.** The gauge pressure in a liquid at a depth of 2.5 m is read to be 28 kPa. What is the gauge pressure in the same liquid at a depth of 10 m?

**A.** 14 kPa **B.** 84 kPa **C.** 56 kPa **D.** 112 kPa

____ **19.** Two identical balls A and B are kicked at an angle of 37° and 53° with the horizontal respectively, with the same initial velocity. Which ball has the maximum horizontal displacement?

**A.** Ball A
**B.** Ball B
**C.** Both balls will have the same maximum horizontal displacement.
**D.** Impossible to determine.

____ **20.** A projectile is fired in such a way that its horizontal range is equal to three times its maximum height. What is the angle of projection?

**A.** 30° **B.** 37° **C.** 53° **D.** 60°

____ **21.** Which motion CANNOT be an example of projectile motion?

**A.** A bullet fired from a gun.
**B.** A bird flying in the air.
**C.** A ball thrown horizontally into the air.
**D.** A javelin thrown by an athlete.

____ **22.** A rifle is aimed horizontally at a target 30 m away. The bullet hits the target 1.8 cm below the aiming point. What is the initial velocity of the bullet? ($g = 10\ m/s^2$)

**A.** 500 m/s **B.** 50 m/s **C.** 600 m/s **D.** 60 m/s

____ **23.** Lenz's law is the result of the law of conservation of:

**A.** mass **B.** charge **C.** energy **D.** momentum

____ **24.** A rescue airplane travelling at 360 km/h horizontally drops a food package from a height of 320 m when it passes over a car driver stranded in the desert. How far from the car driver should the food package be dropped? ($g = 10\ m/s^2$)

**A.** 340 m **B.** 400 m **C.** 680 m **D.** 800 m

____ **25.** At which position in its flight will a ball experience its minimum speed during inclined projection?

**A.** At the beginning **B.** At the end
**C.** At maximum height **D.** The same speed at all positions

____ **26.** A transformer has primary coil with 2000 loops and secondary coil with 1000 loops. If the current in the primary coil is 4 A, then what is the current in the secondary coil?

**A.** 8 A **B.** 16 A **C.** 2 A **D.** 1 A

____ **27.** A wheel has a radius of 20 cm and accelerates from rest to 15 rev/s in 30 s. What is the magnitude of the tangential acceleration of a point at the tip of the wheel?

**A.** $0.6\ m/s^2$ **B.** $0.5\ m/s^2$ **C.** $2.0\ m/s^2$ **D.** $1.2\ m/s^2$

____ **28.** A ball is kicked into the air from the ground at an angle θ with the horizontal. When the ball reaches its highest point, which statement is true?

**A.** Both the velocity and acceleration of the ball are zero.
**B.** Its velocity is perpendicular to its acceleration.
**C.** Its velocity is not zero, but its acceleration is zero.
**D.** Its acceleration depends on the angle at which the ball was thrown.

____ **29.** A ball is kicked from the ground with an initial speed of 25 m/s at an angle of 53° above the horizontal directly toward a wall. The wall is 24 m from the release point of the ball. How long does the ball take to reach the wall? ($\cos 53° = 0.6$)

**A.** 3.2 s **B.** 1.6 s **C.** 0.625 s **D.** 1.3 s

____ **30.** Which of the following cannot be a unit for angular displacement?

**A.** degree **B.** radian **C.** revolution **D.** meter

____ **31.** Consider two particles, A and B, on a flat rotating disk. Particle A is 20 cm and particle B is 40 cm from the center. The disc starts from rest and its angular speed increases to 20 rad/s in 4 s. What is the average angular and linear acceleration for particle B, respectively?

**A.** $5\ rad/s^2$, $1\ m/s^2$ **B.** $10\ rad/s^2$, $2\ m/s^2$
**C.** $5\ rad/s^2$, $2\ m/s^2$ **D.** $10\ rad/s^2$, $1\ m/s^2$

____ **32.** The magnetic flux is maximum when the angle between magnetic field lines and the line perpendicular to the plane of the area is:

**A.** 0° **B.** 90° **C.** 45° **D.** 30°

____ **33.** A long straight wire carrying a current produces a magnetic field of 0.8 T at a distance 0.5 cm from the wire. What is the magnetic field at a distance of 1 cm?

**A.** 0.2 T **B.** 0.4 T **C.** 1.6 T **D.** 3.2 T

____ **34.** The emf induced in a coil can be increased by:

**A.** Increasing the number of turns per unit length in the coil (n)
**B.** Increasing the rate of change of magnetic field strength surrounding the coil per unit time
**C.** Increasing the speed of the relative motion between the coil and the magnet
**D.** All of the above are correct statements.

____ **35.** Three particles are connected by rigid rods of negligible mass lying along the y-axis: 4 kg at y = 3 cm, 2 kg at y = -2 cm, and 3 kg at y = -4 cm. If the system rotates about the x-axis, what is the moment of inertia about the x-axis?

**A.** 36 kg·cm² **B.** 4 kg·cm² **C.** 92 kg·cm² **D.** 20 kg·cm²

____ **36.** Which of the following is true about magnetic field lines?

**A.** They form closed loops.
**B.** They never intersect each other.
**C.** The magnetic field lines are crowded near the pole.
**D.** All are true statements.

____ **37.** A 500 turns coil develops an average induced voltage of 60 V. Over what time interval must a flux change of 0.06 Wb occur to produce such a voltage?

**A.** 5.0 sec **B.** 0.5 sec **C.** 2.0 sec **D.** 0.2 sec

____ **38.** Which one of the following is false about magnets?

**A.** Every magnet has two poles.
**B.** Like poles attract each other.
**C.** A magnet generates a magnetic field.
**D.** The magnetic field is stronger at the poles.

____ **39.** Two objects are attracted to each other by a gravitational force F. If the distance between the objects is doubled, what is the new gravitational force between the objects in terms of F?

**A.** 4 F **B.** 1/4 F **C.** 16 F **D.** 1/16 F

____ **40.** A force of 400 N is applied to a beam at a distance of 5 m from the pivot point at an angle of 30° to the beam. What is the magnitude of the torque which turns the bar around the pivot?

**A.** 2000 Nm **B.** 1733 Nm **C.** 867 Nm **D.** 1000 Nm